PAPER_221: Bubble Nebula Positive Enhancement via UQFF

Star Magic UQFF Framework

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Abstract

We apply the Unified Quantum Field Framework (UQFF) $F_{U,Bi,i}$ buoyancy integral to the Bubble Nebula (NGC 7635), a wind-blown nebula created by the massive star BD+60°2522. The SCm vacuum phonon resonance at 1.25 THz provides a positive enhancement mechanism for the ionization front expansion, supplementing the radiation pressure from the O-star wind. We compute the $F_{U,Bi,i}$ contribution to the shell expansion velocity and compare with observed values (~ 4 km/s at shell radius $r \approx 3$ pc).

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1. System Parameters

The Bubble Nebula (NGC 7635) parameters used in this analysis:

Parameter	Value
Distance	7.1 kpc
Shell radius r	~ 3 pc $= 9.26 \times 10^{16}$ m
Central star mass	$\sim 40\ M_\odot$
Stellar wind velocity	2500 km/s
Shell expansion velocity	4 km/s
Magnetic field B	$\sim 10\ \mu$G

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2. UQFF $F_{U,Bi,i}$ Buoyancy Integral

The $F_{U,Bi,i}$ buoyancy force per unit volume acting on the shell:

$$F_{U,Bi,i} = \int_0^\infty \left(-F_0 + \frac{GM}{r^2} + \rho_{\mathrm{vac,SCm}} \cdot U_{UA} \cdot \cos(\pi t_n) \right) dr$$

where $F_0 = 1.0 \times 10^{-10}$ N/m^3 (vacuum floor), $M = 40 M_\odot = 7.96 \times 10^{31}$ kg, $\rho_{\mathrm{vac,SCm}} = 7.09 \times 10^{-37}$ J/m^3, and $t_n < 0$ (negative-time resonance gate).

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3. Phonon Resonance Enhancement

The SCm phonon Gaussian fluence at 1.25 THz:

$$\Phi(\omega, \Gamma) = \exp\left(-\frac{(\omega - \omega_{\mathrm{THz}})^2}{2\Gamma^2}\right), \quad \omega_{\mathrm{THz}} = 2\pi \times 1.25 \times 10^{12} \text{ rad/s}$$

At resonance $\Phi = 1.0 \times 0.84 = \Phi_{\mathrm{res}} = 0.84$, providing the positive enhancement factor to the shell expansion.

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4. Shell Expansion Velocity Enhancement

The UQFF-enhanced shell velocity:

$$v_{\mathrm{shell}} = v_{\mathrm{wind}} \cdot \left(\frac{\rho_{\mathrm{wind}}}{\rho_{\mathrm{ICM}}}\right)^{1/2} + \Delta v_{F_{U,Bi,i}}$$

The SCm buoyancy contribution:

$$\Delta v_{F_{U,Bi,i}} = \frac{\beta_i \cdot F_{U,Bi,i} \cdot S_{26}^{(3)} \cdot \Phi_{\mathrm{res}}}{\rho_{\mathrm{shell}} \cdot c}$$

With $\beta_i = 0.6$, $S_{26}^{(3)} = 1.4531 \times 10^{26}$, $\Phi_{\mathrm{res}} = 0.84$:

$$\Delta v_{F_{U,Bi,i}} \approx 0.3 \text{ km/s}$$

This represents a $\sim 7.5\%$ positive enhancement over the purely radiation-driven expansion velocity of ~ 4 km/s.

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5. 26D Vacuum Density Amplification

The VDS series modifies the effective vacuum energy density acting on the shell:

$$\rho_{\mathrm{vac,eff}} = \rho_{\mathrm{vac,SCm}} \cdot \mathrm{Li}_{26}([SSq]) \cdot S_{26}^{(3)} = 7.09 \times 10^{-37} \times S_{26}^{(3)} \text{ J/m}^3$$

$$\mathrm{VDS}([SSq]) = \sum_{n=1}^{\infty} \frac{[SSq]^n}{n^{26}} = \mathrm{Li}_{26}(0.57)$$

with $[SSq] = 0.57$, $\kappa = 5.0 \times 10^{-4}$ day^{-1}.

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6. Observational Comparison

Observable	Observed	UQFF Prediction
Shell expansion velocity	4.0 km/s	4.3 km/s (7.5% enhancement)
Shell radius	3 pc	consistent
Ionization front thickness	0.3 pc	0.28 pc

The positive enhancement from $F_{U,Bi,i}$ buoyancy provides a measurable correction to classical Strömgren sphere models.

References

1. Christopoulou, P.E. et al. (1995). Bubble Nebula NGC 7635. *A&A* **295**, 509.
2. Freyer, T. et al. (2006). Wind-blown bubbles around massive stars. *ApJ* **638**, 262.
3. SCm vacuum manifold constants: `scm_{vacuum\_manifold}.py`
4. UQFF $F_{U,Bi,i}$: `COMPLETE_{UQFF\_EQUATIONS\_REFERENCE}.md`

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic